YANCY ONG YONG SIN

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linkedin.com/in/yongsinong | Engineering Portfolio

Mechatronics Engineering student with hands-on experience in PLC programming, control systems, embedded development, and mechanical design. Experienced in developing automation logic, integrating multiple subsystems, and designing HMI interfaces. Competition-winning engineering background with strong problem-solving and collaborative project experience.

EDUCATION

Bachelor of Engineering (Honours) – Mechatronics
Monash University, Clayton
Expected Graduation: July 2027

Visa Status: Student visa subclass 500; eligible to apply for graduate visa subclass 485

Relevant Areas:
Embedded Systems, Control Systems, Engineering Mechanics, Manufacturing, Intelligent Systems

TECHNICAL SKILLS

Programming & Tools: Python, C/C++, Java, MATLAB, Microsoft Excel, Outlook

Embedded & Electronics: STM32, Arduino, circuit prototyping, sensor integration, hardware debugging

Control & Automation: PLC (Ladder Logic), pneumatic systems, sequencing logic design

CAD & Engineering Design: SolidWorks (CSWA – Full Marks), AutoCAD, engineering drawings

ENGINEERING PROJECTS & ACHIEVEMENTS

Warman Competition Robot — Champion, Monash Malaysia

- Worked in a team to design, prototype, and present a competition robot for the Warman Design and Build Competition.
- Contributed to concept generation, CAD modelling, technical documentation, and iterative design refinement.
- Proposed a feasible mechanical design approach that contributed to the team achieving **1st Place at Monash Malaysia**.

PLC-Controlled Pneumatic Can Crushing Machine

- Designed PLC ladder logic for an automated pneumatic can crushing system involving pneumatic actuator control, conveyor movement, actuator sequencing, and emergency stop logic.
- Assisted with electrical panel wiring, commissioning, and troubleshooting.
- Achieved approximately **70% volume reduction per cycle**.

Drone-Assisted Autonomous High-Rise Facade Cleaning Robot

- Proposed the initial system concept and modular architecture, defining movement, cleaning, drone, safety and communication subsystems.
- Designed and implemented movement control using CODESYS PLC Ladder Logic, including finite state machines (FSMs), suction interlocks, actuator sequencing and modular Function Blocks.
- Integrated movement, cleaning, drone and safety subsystems into a coordinated PLC simulation, incorporating operational feedback, safety interlocks and fault handling.
- Developed a multi-level Human–Machine Interface (HMI) for normal operation, engineering diagnostics and simulation testing.

FPGA-based embedded camera project

- Designed and implemented an FPGA-based embedded camera system for image capture and digital processing.
- Worked with hardware-level logic, timing constraints, camera interface debugging, and system integration.
- Strengthened skills in embedded hardware, real-time system behaviour, and low-level troubleshooting.

Breathing Strain Sensor & Thermistor Measurement Projects

- Developed sensor-based measurement systems using strain sensing and thermistor-based temperature detection.
- Applied calibration, data collection, signal interpretation, and noise consideration to improve measurement reliability.

SmartFab: AI-Driven Voice-to-Product Manufacturing Concept

- Developed a competition concept for an AI-driven manufacturing system that converts natural language product requirements into a manufacturable workflow.
- Proposed a modular prototype integrating conversational AI, CAD generation, robotic material handling, and automated manufacturing demonstration.
- Contributed to feasibility analysis, implementation planning, and prototype simplification for a student engineering team.

Monash Motorsport Recruit

- Participated as a Monash Motorsport recruit, gaining exposure to student-led motorsport engineering, design workflows, coding, documentation, and testing processes.

LEADERSHIP EXPERIENCE

Head of Events — Monash Foundation Year Student Council

- Led event planning, logistics, scheduling, and cross-team communication for student activities.
- Developed leadership, organisation, communication, and problem-solving skills through event coordination.

Team Leader — Engineering Design Unit

- Led team coordination, task allocation, and project progress tracking for an engineering design project.
- Supported final technical presentation delivery, contributing to a **Best Presentation Award**.

WORKING EXPERIENCE

Design Engineer — Wah Lam Building Materials Limited, Hong Kong

Remote

- Supported 2D CAD drafting, proposal design, and project documentation tasks under tight deadlines.
- Completed urgent design and paperwork tasks while maintaining clear communication in a remote working environment.

Front of House Team Member — Angie Dumpling

- Provided customer service in a fast-paced restaurant environment while supporting smooth daily operations.

Event Crew — Proton, Kuala Lumpur International Mobility Show

- Engaged with visitors to promote Proton vehicles and explain product information.
- Supported event operations through customer communication, teamwork, and professional presentation.